

Week 9 – Backend Testing & Error Handling Enhancement

Summary:

This week focused on **extensive backend testing** and improving **error handling** to ensure reliable execution of multi-language programs under different scenarios.

Key Activities:

- Tested backend with **large and complex programs** in Python, C, C++, and Java.
- Identified edge cases that could cause execution errors and implemented **robust error handling**.
- Logged **compilation and runtime errors** systematically for review.
- Verified Docker container isolation for multi-language execution to prevent conflicts.

Technologies Used: Python, FastAPI, Docker, GCC, g++, JDK, WSL2

Outcome:

Backend can now **handle larger programs**, manage errors efficiently, and maintain stable execution across all supported languages.